



SEED DISPENSARY ROVER

An Automated Seed Sowing Robot for Smart Agriculture

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SEED DESPENSER ROVER

Traditional seed sowing methods require significant human effort and time. Farmers often face difficulties in maintaining proper seed spacing and uniform distribution, leading to reduced productivity and wastage of seeds.

Challenges:

- Manual labor requirement
- Uneven seed distribution
- Increased time consumption
- Seed wastage
- Higher farming costs

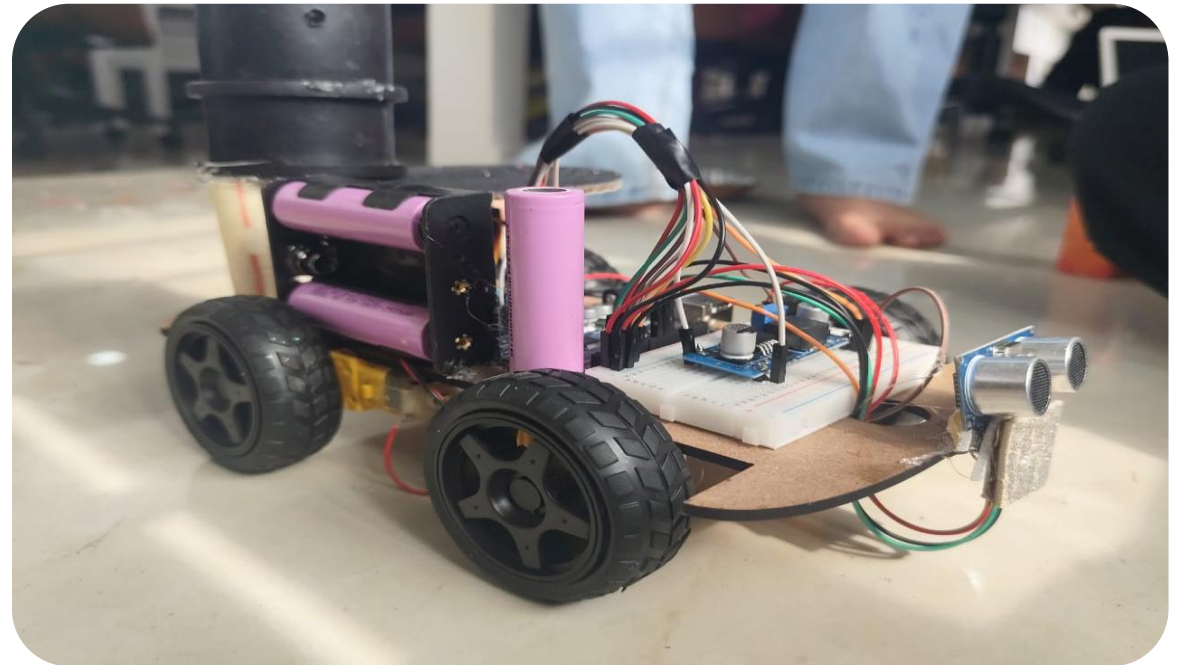


PROPOSED SOLUTION

The Seed Dispensary Rover is an automated robotic system designed to sow seeds efficiently while moving through agricultural land.

Features

- Automatic seed dispensing
- Obstacle detection
- Autonomous movement
- Uniform seed distribution
- Low-cost and easy-to-use design



Objectives

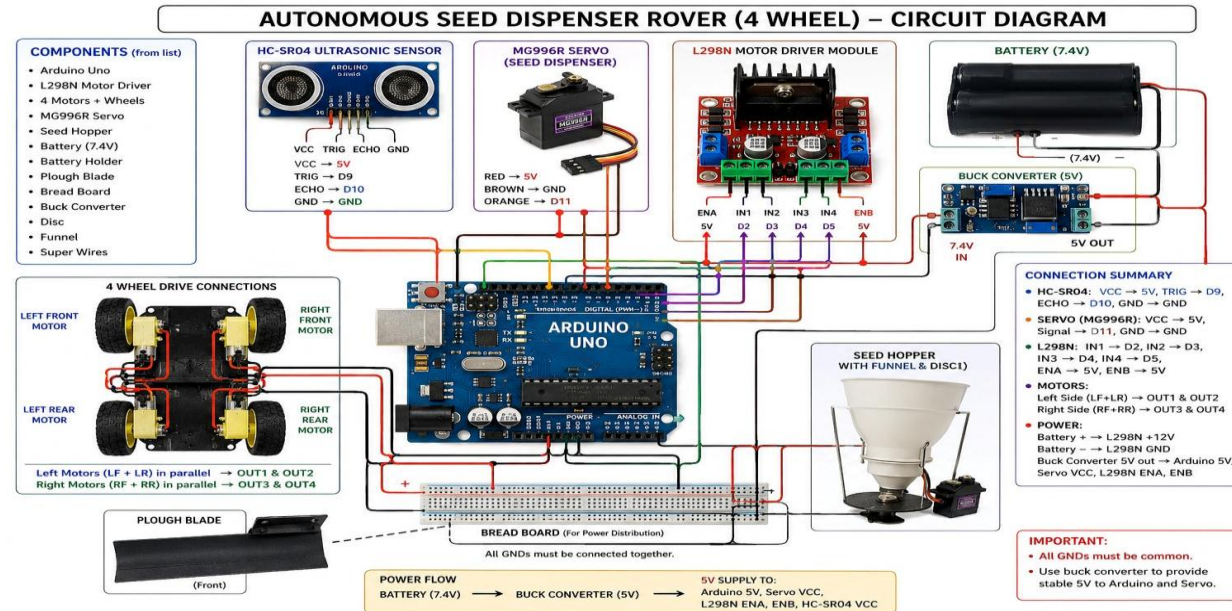
- Automate the seed sowing process.
- Reduce human effort in farming.
- Improve seed placement accuracy.
- Minimize seed wastage.
- Increase agricultural productivity.
- Provide a cost-effective solution for farmers.

Components Used

- Arduino Uno
- SG90 Servo Motor
- DC Motors
- Motor Driver Module (L298N)
- Ultrasonic Sensor (HC-SR04)
- Chassis with Wheels
- Seed Container
- Battery Pack
- Connecting Wires

CIRCUIT DIAGRAM

Arduino Uno as the main controller. The ultrasonic sensor detects obstacles, the motor driver controls the DC motors for movement, and the SG90 servo motor dispenses seeds at regular intervals.



Working Principle

- The rover moves forward automatically.
- Ultrasonic sensor continuously detects obstacles.
- If an obstacle is detected, the rover changes its direction.
- The servo motor rotates periodically.
- Seeds are dispensed through the seed container outlet.
- Seeds are placed at regular intervals during movement.
- **Result:**
- Uniform seed sowing with minimal human intervention.

System Architecture

Start



Move Forward



Check Distance



Obstacle?

├ Yes → Stop → Reverse → Turn → Continue



└ No → Continue Forward



Dispense Seed



Repeat

Advantages

- Reduces manual labor
- Saves time
- Uniform seed placement
- Low maintenance
- Cost-effective
- Easy operation
- Suitable for small-scale farms

Future Scope

- GPS-based navigation
- Mobile application control
- Soil moisture monitoring
- IoT-based data collection
- Solar-powered operation
- AI-based route optimization



Conclusion

- The Seed Dispensary Rover provides an efficient and affordable solution for modern agriculture. By automating the seed sowing process, it reduces labor requirements, improves seed distribution accuracy, and increases farming efficiency. This project demonstrates how robotics can contribute to smart farming practices.

THANK YOU